มหาวิทยาลัยมหิดล
Mahidol University

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ITDS261 Introduction to Software Engineering

Lecture 1 – Software Engineering Foundation & Software Project Management

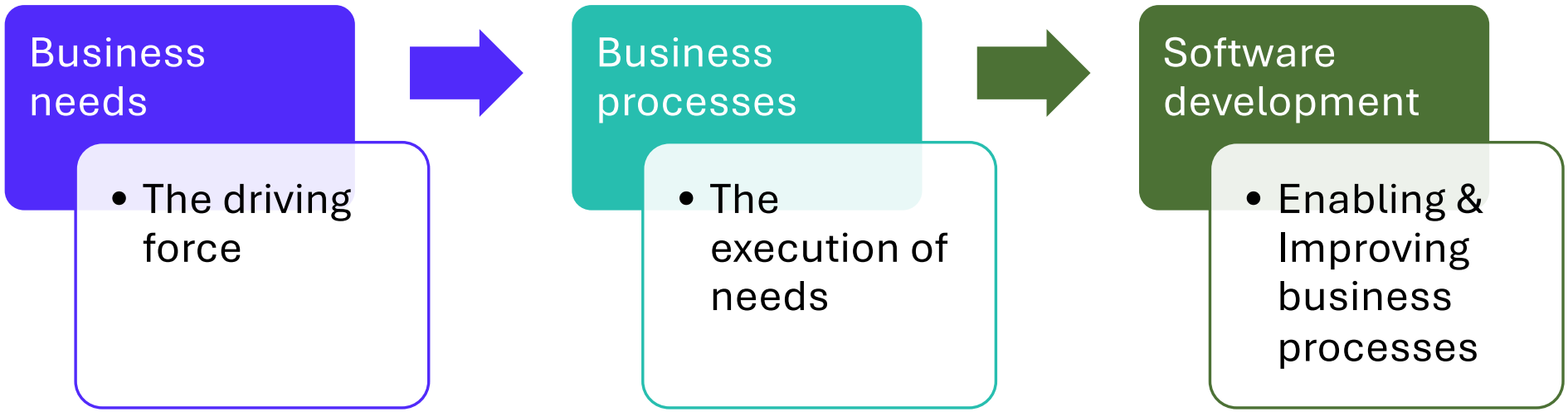
Photo: https://www.flickr.com/photos/ninara/38707088530

Topics

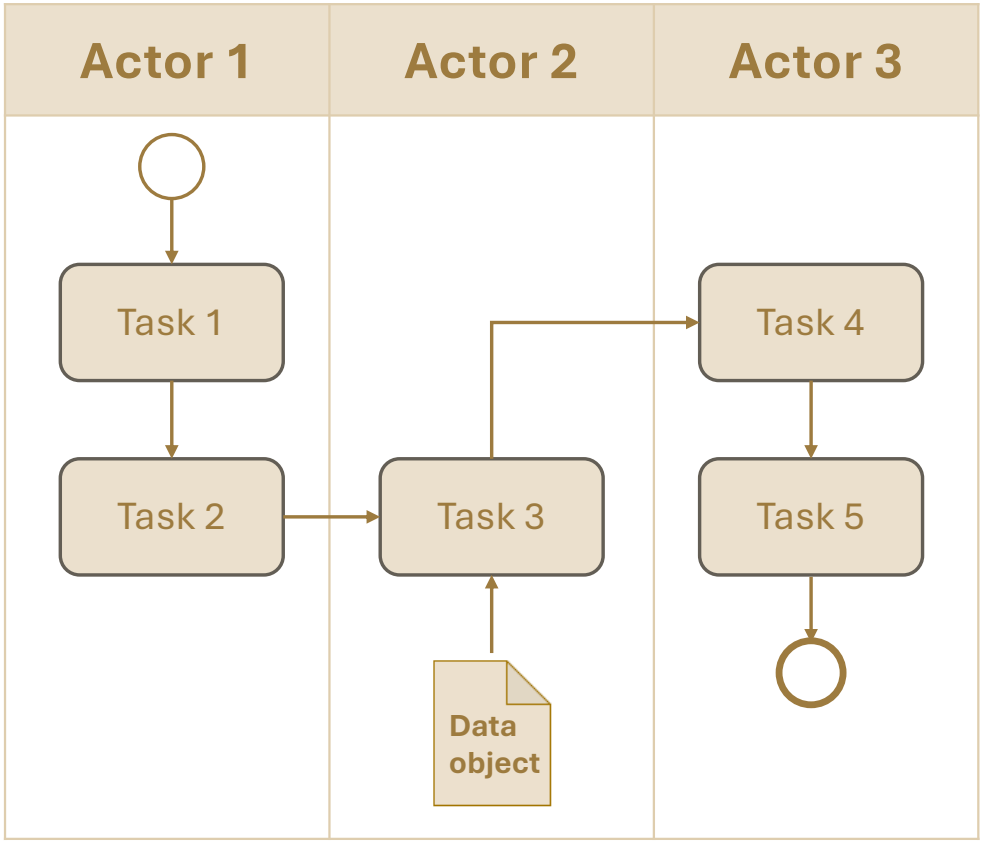
- Software Engineering Foundation
  - Journey to software development
  - Software and its components
  - Software engineering foundation
- Software Project Management
  - Project management overview
  - Project attributes & constraints
  - Project management process
  - Project management framework
  - Sample of project management techniques

Software Engineering Foundation

Business Needs to Software Development



Business Processes



- Key focus:
  - How to get work done?
- Components:
  - Actors
  - Tasks or activities
  - Rules
  - Inputs and outputs
- Characteristics:
  - Can be modelled using BPMN
  - May or may not need a system to support

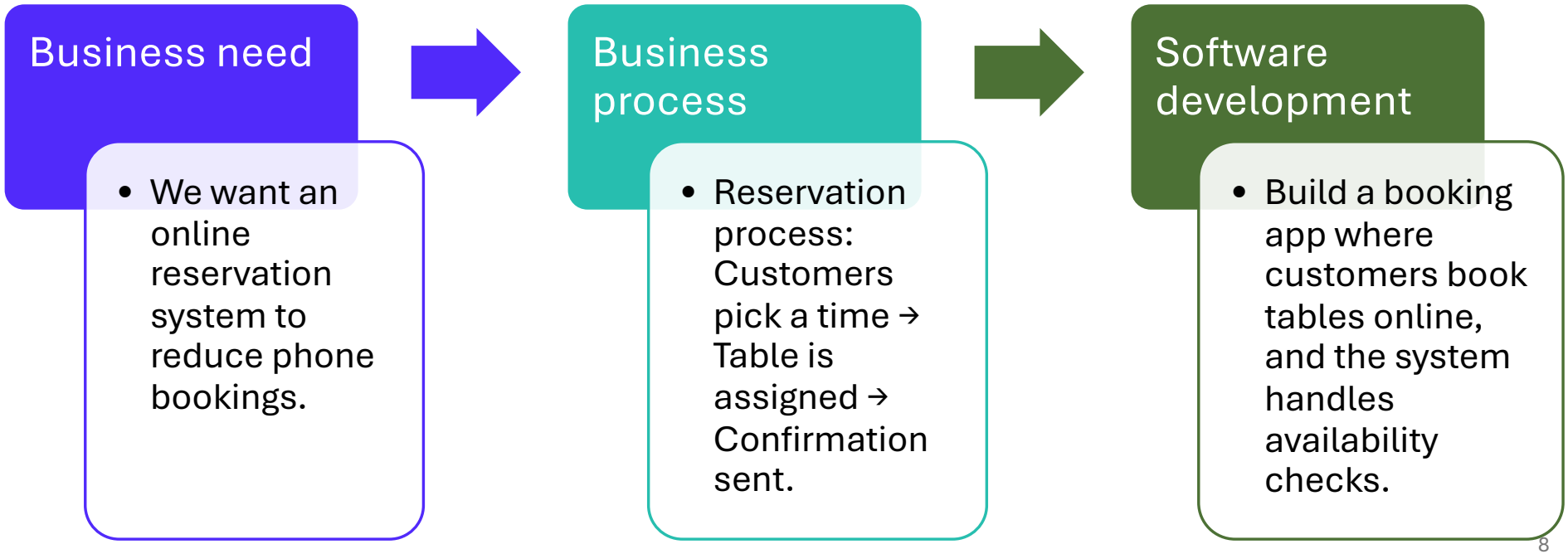


Software Development

- To make business processes more efficient, accurate and scalable.
- To ensure that business processes are digitised, automated and improved based on business needs.

Bringing It All Together

Imagine a **manual restaurant reservation system** where customers call, and staff manually record bookings. The restaurant has a **business goal** to reduce human errors and improve efficiency.



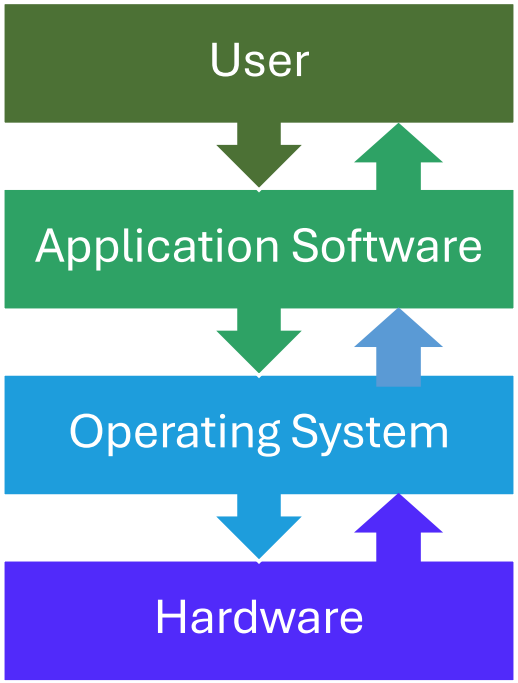
What is software?

Software and Its Components

1) A collection of *instructions* that provide features, function and performance
2) Data structures that enable the programs to manipulate information
3) A descriptive information *(document)* that describes the operation and use of the programs

Reference: Pressman, R.S. (2015) *Software engineering: A practitioner's approach*. New York, NY: McGraw-Hill Education.

Interaction Hierarchy



Reference: Pressman, R.S. (2015) *Software engineering: A practitioner's approach*. New York, NY: McGraw-Hill Education.

Characteristics of Software

- Need to be developed or engineered, not manufactured
- Intangible
- Labour intensive
- Difficult to make changes
- Does not wear out
- Can be obsolete

Examples of Software Obsolescence



Windows XP
(obsolete in 2014)



Windows 7
(obsolete in 2023)

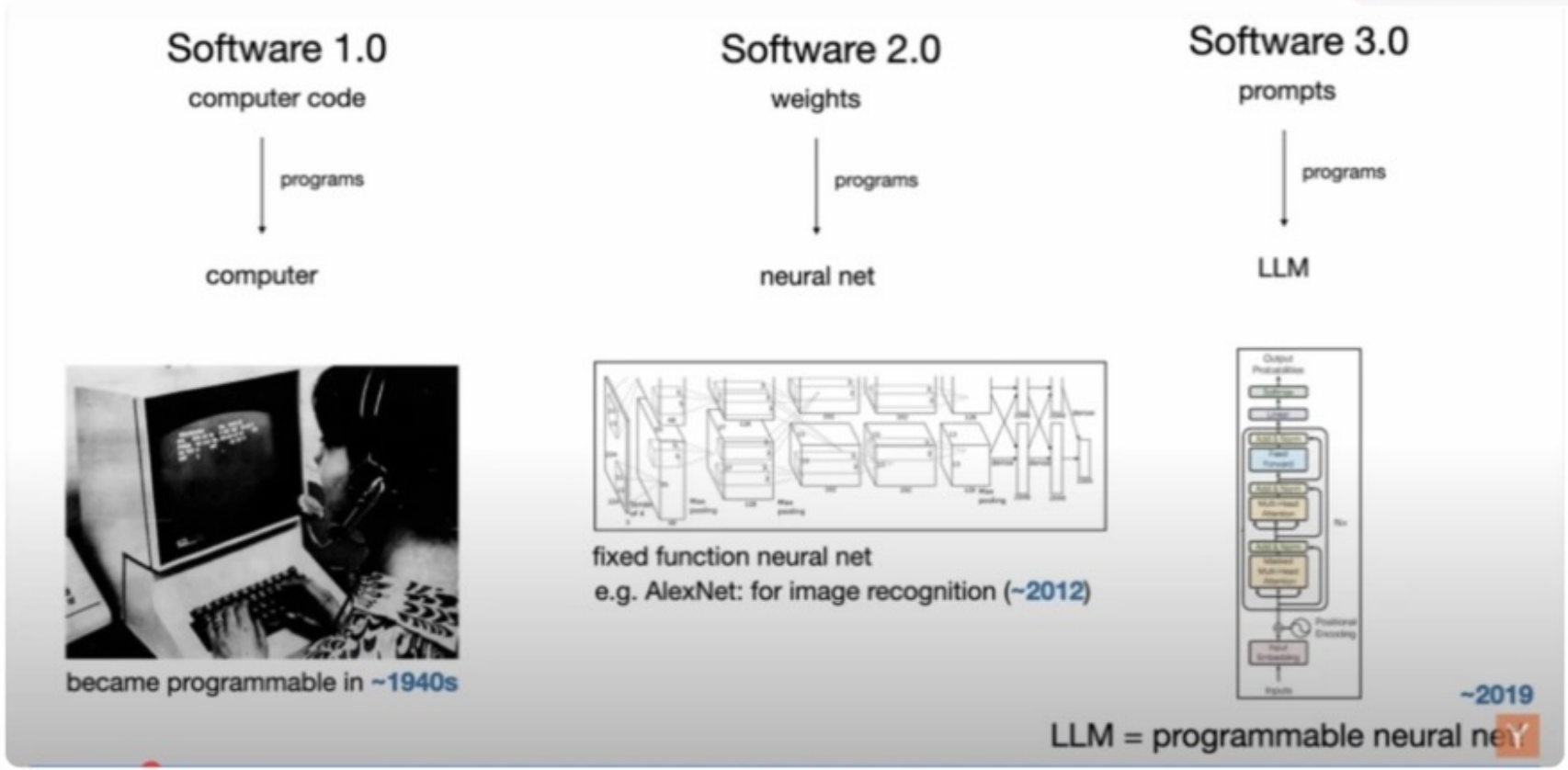
Types of Software Applications

- Software applications can be classified into several types based on their purpose and functionality.

Productivity	Communication	Media players / editors	Education	Business
• Microsoft Word / Excel / PowerPoint • Canva • Notion	• Zoom • Slack • Microsoft Teams • Webex	• VLC Media Player • Adobe Photoshop / Premier Pro • Audacity	• Duolingo • Coursera • EdX • Quizlet • Lightbot	• QuickBooks (accounting) • Salesforce (CRM) • SAP (ERP)

Evolution of Programming





Source: https://youtu.be/LCEmiRjPEtQ

“Software is eating the world”

- Marc Andreessen



Reference: https://a16z.com/why-software-is-eating-the-world/



Reference: https://www.solzit.com/the-future-of-software-development-7-trends-to-follow-in-2026/

What is software engineering (SE)?

Software Engineering (SE)

A process of *solving customers' problems* by the *systematic development and evolution* of *large, high-quality software systems* within *cost, time and other constraints*.

- Timothy C. Lethbridge



Graphic was produced by AI

Key Points

- Solving customers’ problems >> our goal
- Systematic development and evolution >> engineering
- Large, high-quality software systems >> very complex
- Cost, time and other constraints >> our limitation

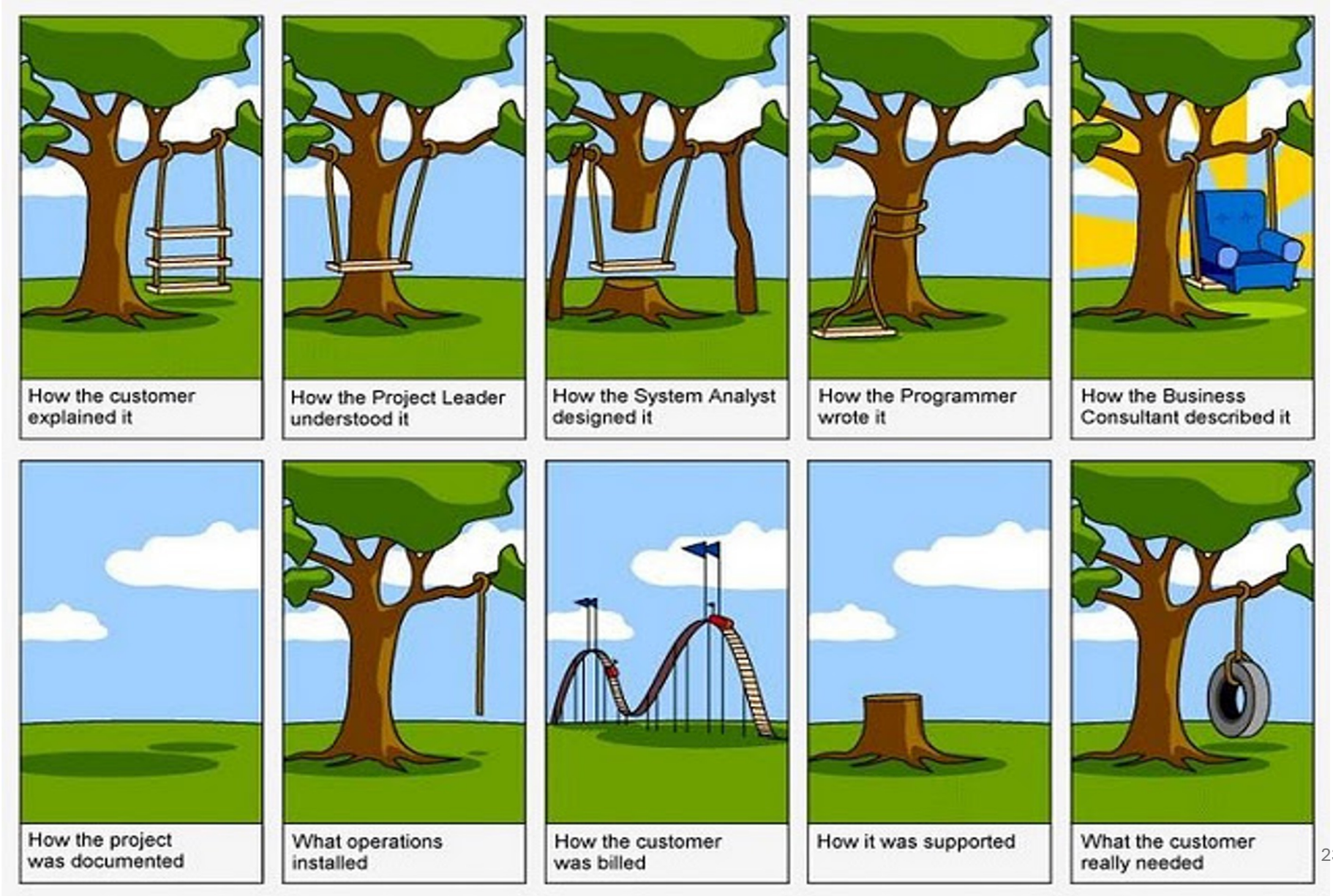
Aspects in Software Engineering

To *develop and support* a software system, these aspects are considered:

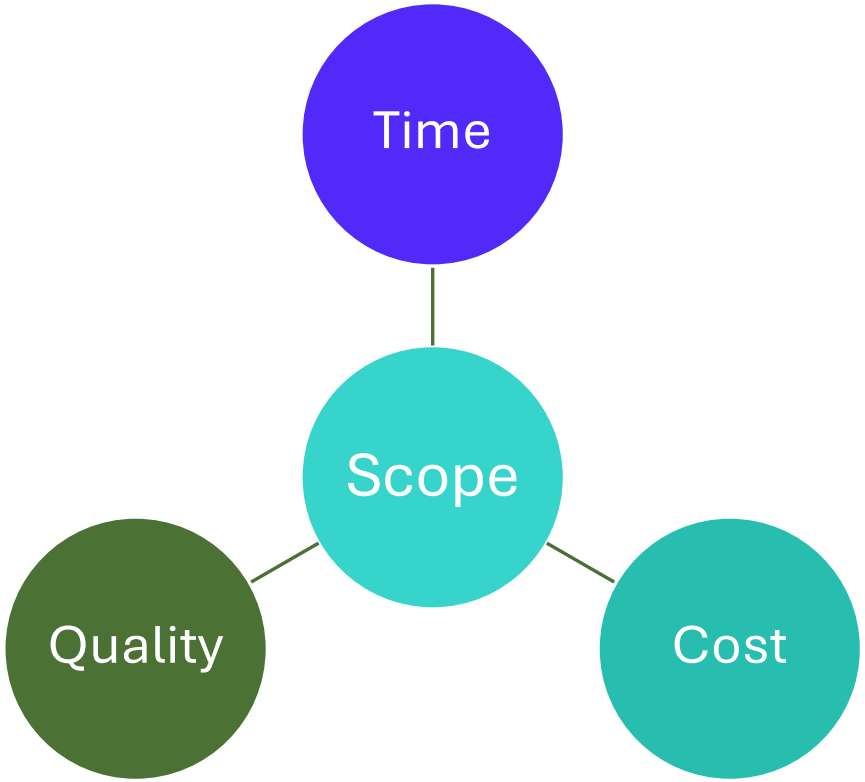
1. Technical and business processes
2. Specific methodologies and techniques
3. Product characterization and metrics for measurements
4. People skills and teamwork
5. Project coordination and management

Reference: Essentials of Software Engineering, Tsui and Karam

Why do we need “software engineering”?


How the customer explained it
How the Project Leader understood it
How the System Analyst designed it
How the Programmer wrote it
How the Business Consultant described it
How the project was documented
What operations installed
How the customer was billed
How it was supported
What the customer really needed

Software Projects



A software project needs:

- Clear objectives and goals
- Clear requirements
- Stakeholder involvement
- Executive management support
- Realistic planning
- Good management

Factors to Success or Failure

- Key success factors:
  - User involvement
  - Executive management support
  - Clear requirement statements
  - Proper planning
- Top failure reasons:
  - Lack of user input
  - Incomplete requirements
  - Changing requirements

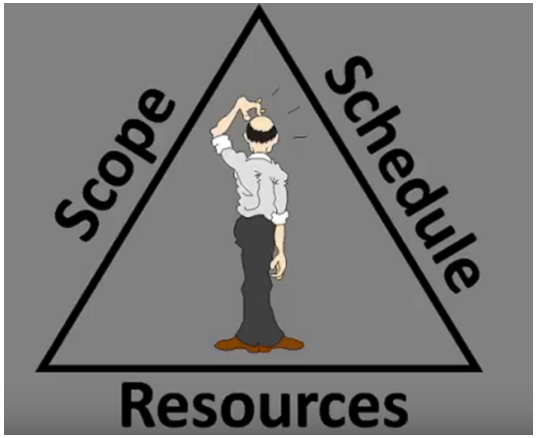


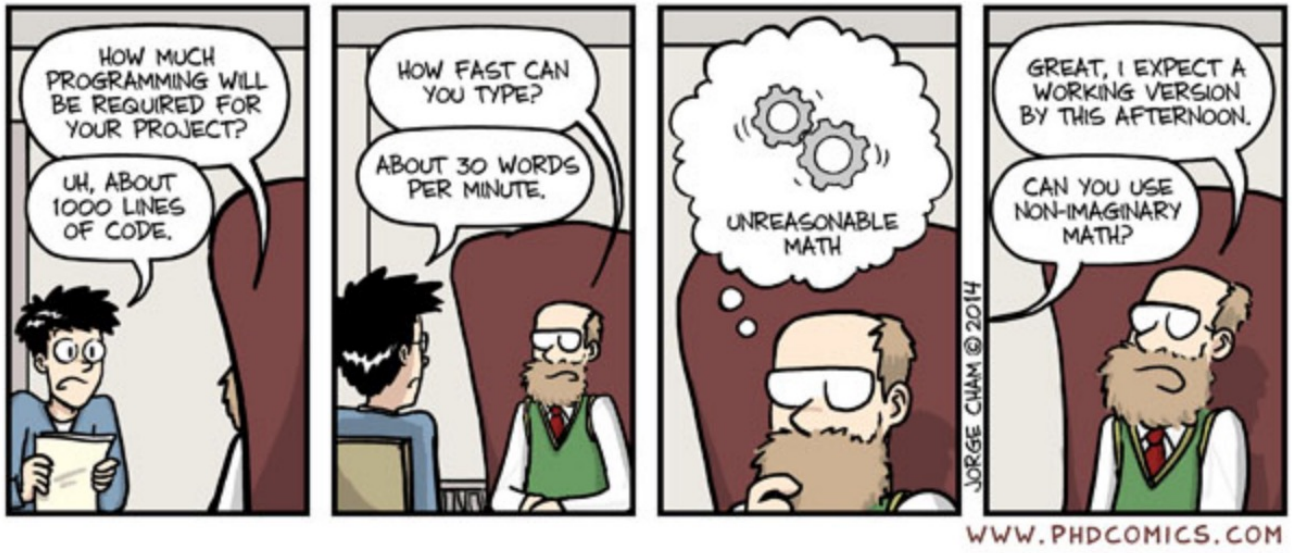
Code errors: 38.33%
Design errors: 24.17%
Documentation errors: 13.33%
Requirements errors: 12.50%
Bad-fix errors: 11.67%

(Chaos & Capers Jones Report)

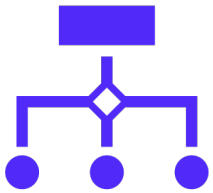
Estimation

- How much effort is required ?
- Cost
- Time / Scheduling





Requirements



- Requirements – define and qualify system
  - Defined by client, with help from engineer
  - Functional – define what must be done
  - Non-Functional – qualify the functional ones



- Design constraints
  - On design or implementation
  - Programming language, platforms etc.

Client side

Roles in a Software Project

Customers

- Define the business needs
- Fund the project
- Approve the final product

End users

- Use the final software
- Provide feedback on usability and functionality

Development side

Roles in a Software Project (Cont.)

Project Manager (PM)

- Plans and manages the project
- Defines scope
- Schedules tasks,
- Allocates resources
- Monitors progress
- Manages risks
- Coordinates stakeholders
- Ensures the project is completed on time and within budget

Business Analyst (BA)

- Elicits, analyses and validates business, functional and non-functional requirements
- Produce requirement specification documents
- Acts as the bridge between customers/end users and the development team

System Analyst (SA)

- Translate business, functional and non-functional requirements into technical specifications
- Designs the overall software architecture, selects technologies and frameworks
- Defines coding standards and architectural patterns

Development side

Roles in a Software Project (Cont.)

Software Developer

- Implements software features
- Writes and maintains source code
- Performs unit testing
- Fixes defects
- Reviews code
- Integrates software components

UX/UI Designer

- Conducts user research
- Creates user personas, wireframes, mockups, and prototypes
- Designs user interfaces, and improves usability and user experience

Quality Assurance (QA) Engineer / Tester

- Develops test plans and test cases
- Performs functional and non-functional testing
- Reports defects
- Verifies bug fixes
- Ensures software quality

Technical Issues in Software Development

Process and Methodology

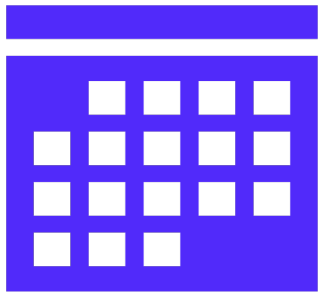
- Choice of process
- Choice of methodologies
- Choice of tools to support the process

Technical Issues in Software Development

Technology and tools choices

- Development platform
- Development language
- Database
- Network
- Configuration management

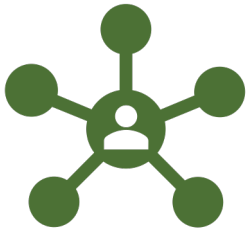
Non-technical Issues in Software Development



Effort Estimation and Scheduling

- Needs to consider and estimate more items
- Needs to coordinate more items in terms of pre-requisites and co-requisites
- Needs to consider more potentials of risks and variations

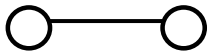
Non-technical Issues in Software Development



Assignments and Communications

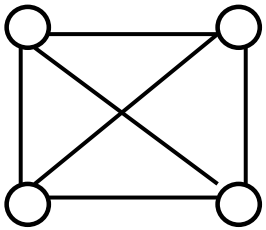
- More people with an increased variety of skills
- More communications among the people
- More errors and modifications

Adding more people to solve the problem?



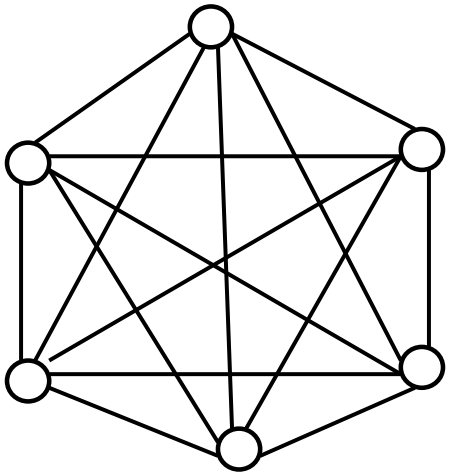
2 people:

1 path



4 people:

possibly 6 paths



6 people:

increase to potentially 15 paths

Increase in amount of communications as # of people increases
Also, an increase in the probability of error

Developing and Supporting a System

What is a system?

- Is a single program a system?
- How many programs must be there?
- Does it have to involve hardware, programs, business process, and others?

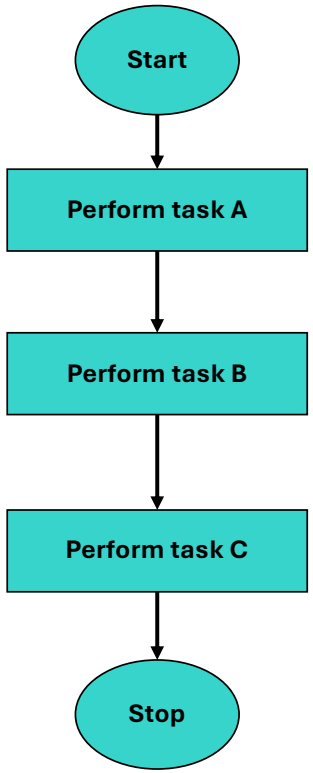
Is there a difference between developing and supporting

- a) a single program and
- b) a system ?

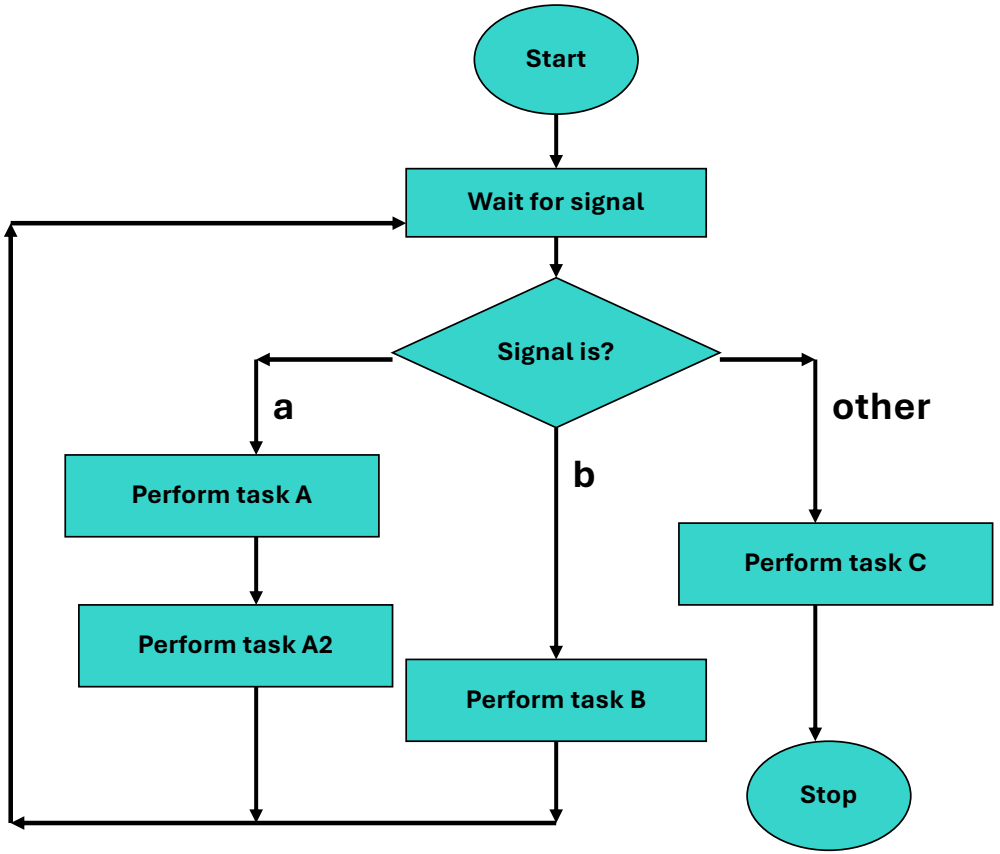
Size and Complexity

- Breadth issue (more in numbers):
  - Major functions
  - Features within each major function
  - Interfaces among features and functions
  - Interfaces to “external” systems
  - Number of simultaneous users and of total users
- Depth issue (linkages)
  - Amount of Nesting
  - Number of “chained” relationships
- How does depth-and-breadth influence complexity?

Size and Complexity



A



B

Implementation

- Coding conventions
  - Cover file organization, indentation, comments, declarations, statements, white space, naming conventions, programming practices, programming principles, programming rules of thumb, architectural best practices, etc.
  - Be consistent
  - Choose names carefully
- Test before using
- Know the libraries
- Do code reviews

Testing

- Acceptance testing: if it fails, client rejects program
- Unit testing: by programmer, on each piece
- Black-box: assume no knowledge of code, check the data validation
- White-box: test the code path as written

Supporting a System

Pre-release education and preparation

- Number of expected users
- Number of known problems and expected quality
- Amount of user and support personnel training
- Number of fix and maintenance cycle

Post-release user and customer support

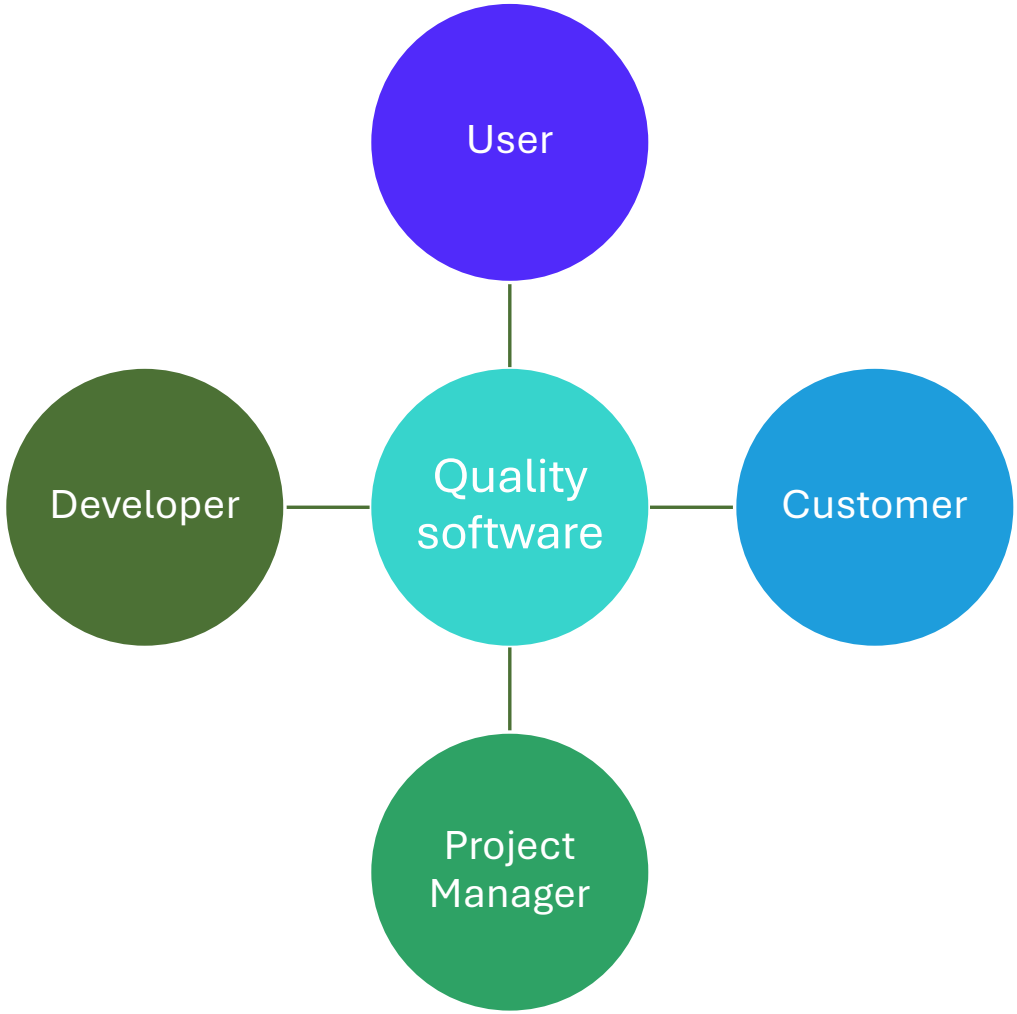
- Call center and problem resolutions
- Major problem fixes and code changes
- Functional modifications and enhancements

Software Quality

- Examples of software quality indicators
  - Usability
  - Efficiency
  - Testability
  - Maintainability
  - Reusability

How can we measure those things?

What does software quality means to different stakeholders?



Summary

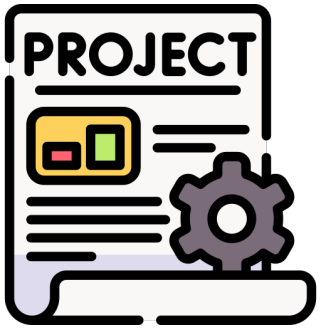
Term	Definition
Software	Programs, data, and documentation that together form a usable system
Software Engineering	The disciplined application of engineering principles to software development
Software Process	The structured set of activities required to develop a software system
Software Product	The artefact produced by the software process — the deployed system and its documentation

Reference: https://book.agentic-swe.dev/chapter_1.html#computer-science-vs-software-engineering

Software Project Management

Software Project Management

- An umbrella activity within software engineering
- Begins before any technical activity
- Continues throughout software development process





The Management Spectrum (4P's)



Problem
- Objectives
- Scope
- Alternatives
- Technical and management constraints

People
- Roles



Project





Process
- Tasks
- Milestones
- Deliverables
- QA

What is a Project?

- A project is “a temporary endeavour undertaken to create a unique product, service, or result” (PMBOK® Guide, Sixth Edition, 2017)
- Operations are work done to sustain the business
- Projects end when their objectives have been reached, or the project has been terminated



Project Attributes

Who is taking care of a project?

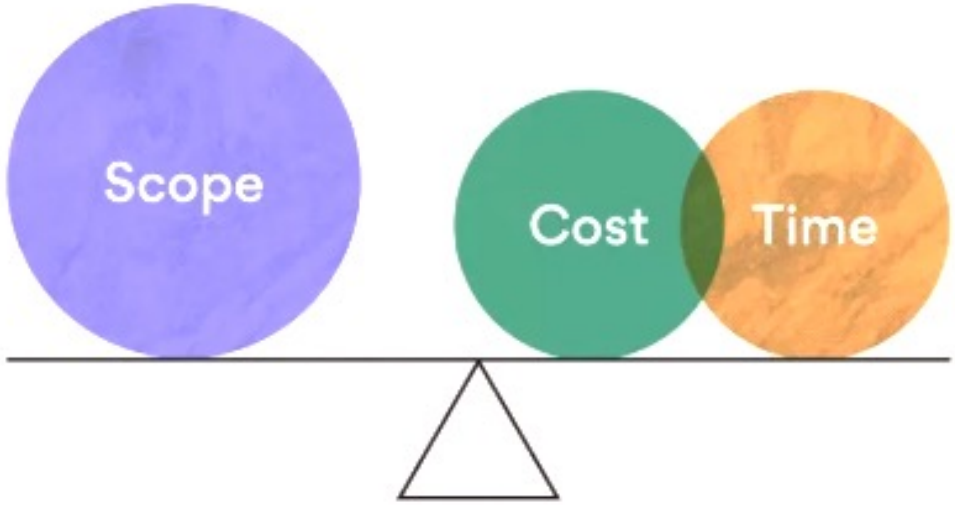
- Has a unique purpose
- Temporary
- Drive change
- Enable value creation
- Require resources
- Has customer or sponsor
- Involves uncertainty

Project Constraints



https://asana.com/resources/project-constraints

Balancing Project Constraints



https://asana.com/resources/project-constraints

What is Project Management?

- The application of knowledge, skills, tools and techniques to project activities to meet project requirements
- Project managers strive to meet the triple constraint (project scope, time, and cost goals) and also facilitate the entire process to meet the needs and expectations of project stakeholders



Project Management Process

- Why do we need project management?
- Why can't we just follow one of the software development process and be left alone?

All projects – small or large – need project management because all projects **need some degree of:**

- planning
- organizing
- status monitoring
- adjustment

Project Management Framework

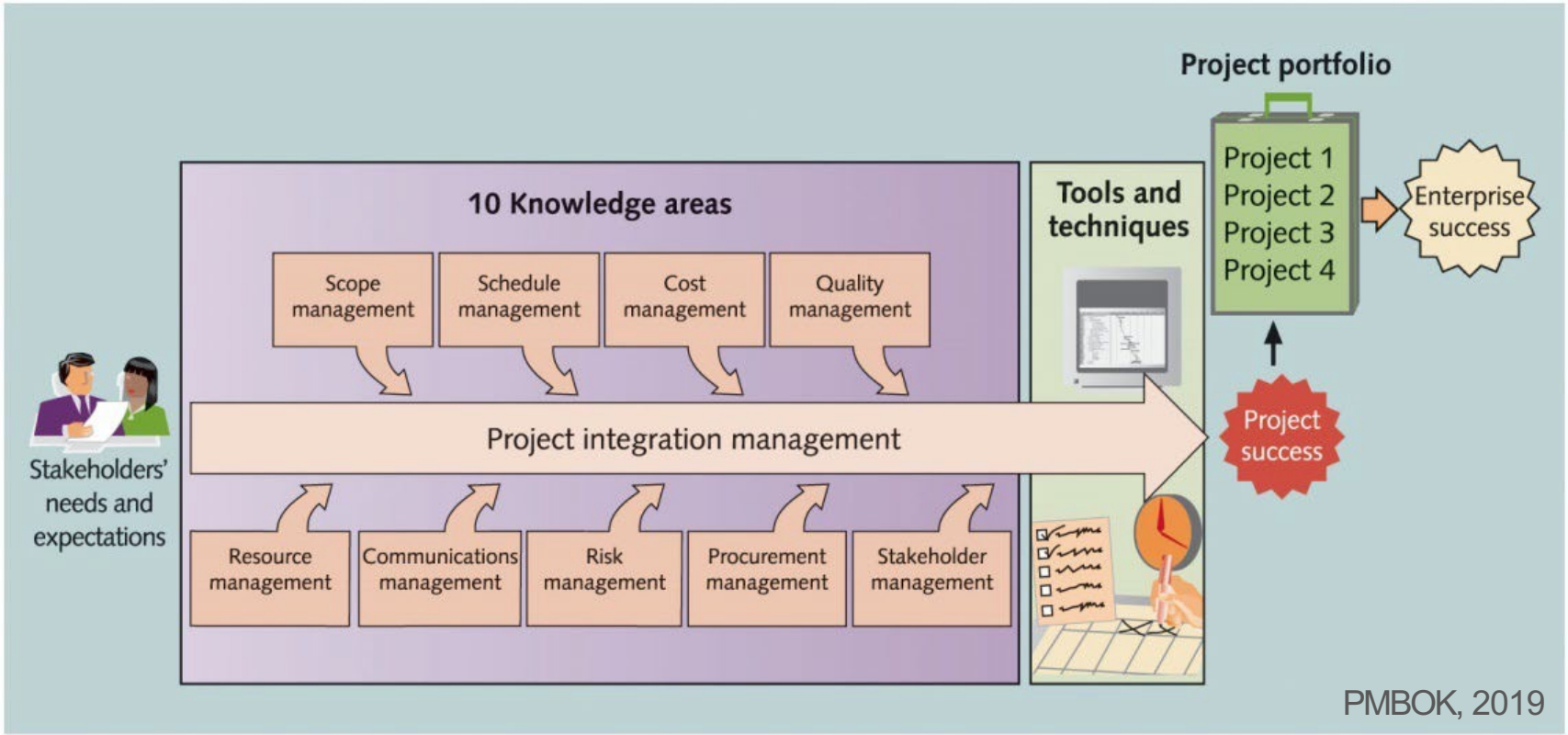


FIGURE 1-2 Project management framework

Project Management Knowledge Areas

- Describe key competencies both knowledge and skills that project managers must develop

Scope	Schedule	Cost	Quality	Resource
Communications	Risk	Procurement	Stakeholder	Project integration management

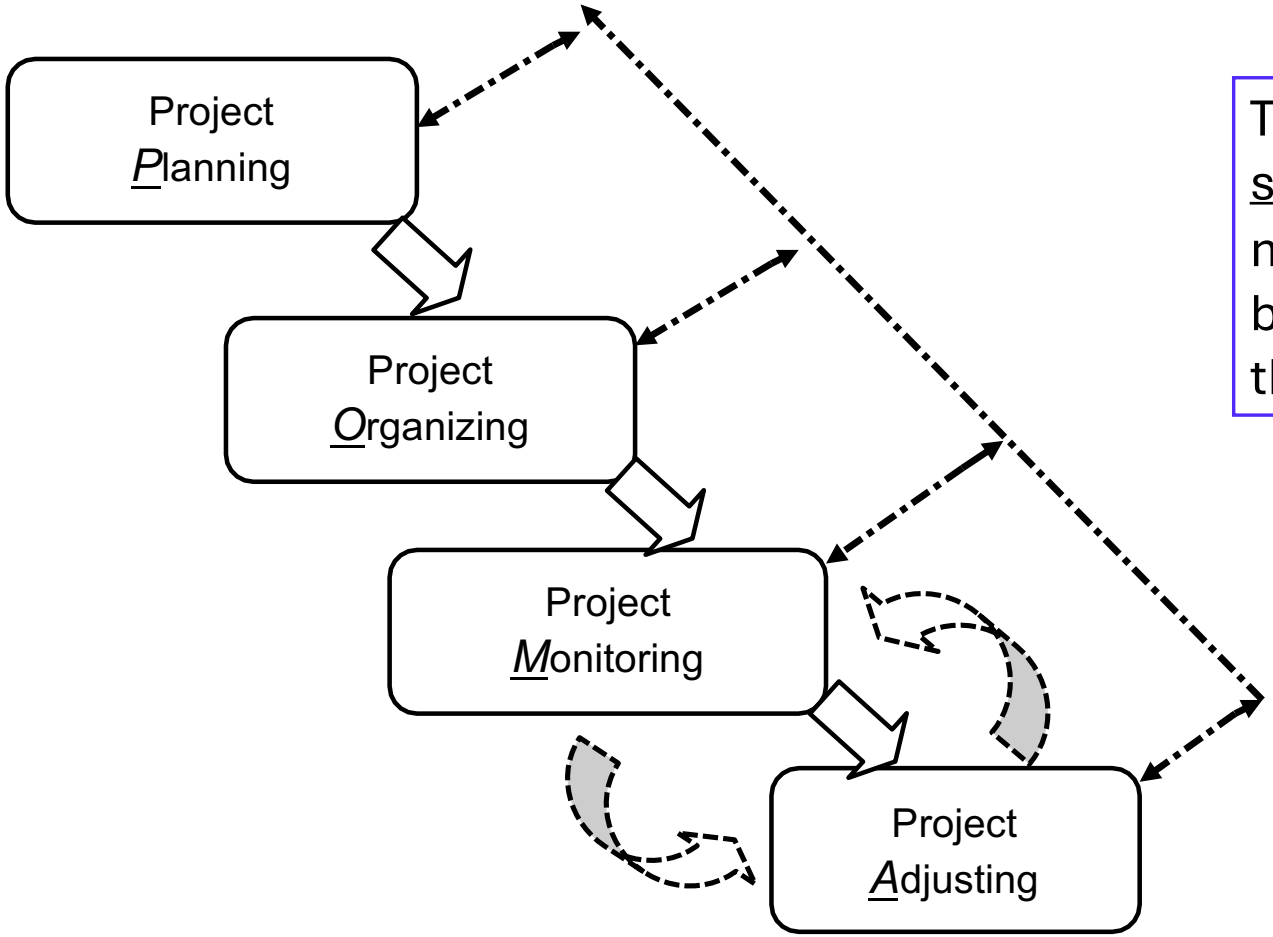
The Role of Project Managers

- Work closely with the other stakeholders on a project, especially the sponsor and project team
- Be familiar with the 10 project management knowledge areas
- Can use various tools and techniques related to project management

Project Stakeholders

- Stakeholders are the people involved in or affected by project activities, can be:
  - Project sponsor or customer
  - Banks and other financial institutions
  - Project manager
  - Project team: technical and non-technical
  - Support staff
  - Suppliers

Software Project Management Process (POMA)



This process looks sequential at the macro level, but may be very iterative at the micro level.

Planning (POMA)

Understand the requirements of the project

Define and establish measurable goals for the project

Estimate work effort

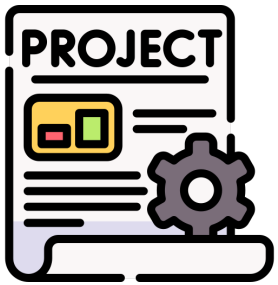
Estimate schedule

Estimate the needed resources

Determine the project resource allocations

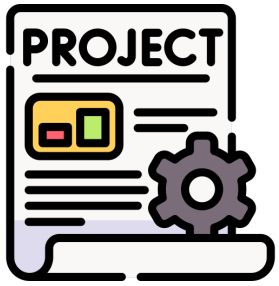
(people, process, tools, and facilities)

Identify and analyze the project risks



Project Plan

Organizing (P**O**MA)



Project Plan

Design organization structure

Hire human resources

Complete required education and training

Initiate tracking mechanisms

- Risk tracking and mitigation
- Project goal monitoring

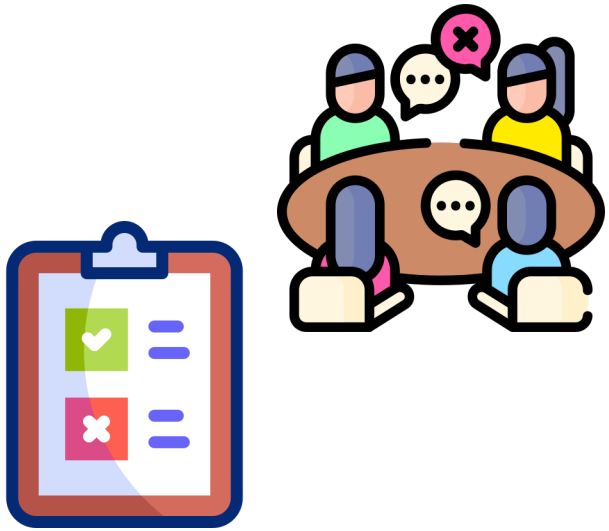
Monitoring (POMA)

- Once the project is organized and set into motion, it needs to be regularly tracked to ensure that it is headed in the right direction.
- 3 main components of project monitoring:

Project status information collection

Analysis and evaluation of collected information

Presentation and communication of the project status



Adjusting (POM**A**)

- It is highly unlikely that a software project progresses with no problem.
- As soon as the project status suggests potential problem, we must not be afraid to make changes.
- 3 main areas of adjustments are (or combinations of)
  - Resources
  - Schedule
  - Project content



Software Project Management Process

is not the same as

Software Development Process or

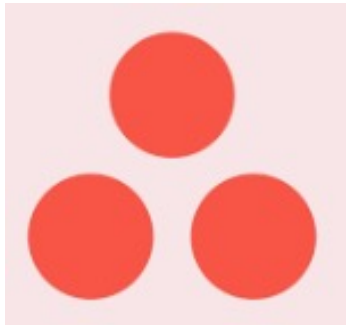
Software Life Cycle

Project Management Tools and Techniques

- Project management tools and techniques assist project managers and their teams in various aspects of project management

- **Examples:**
  - Project charter, scope statement, and WBS (scope)
  - Gantt charts, network diagrams, critical path analysis
  - Cost estimates and earned value management (cost)

Let’s Guess!







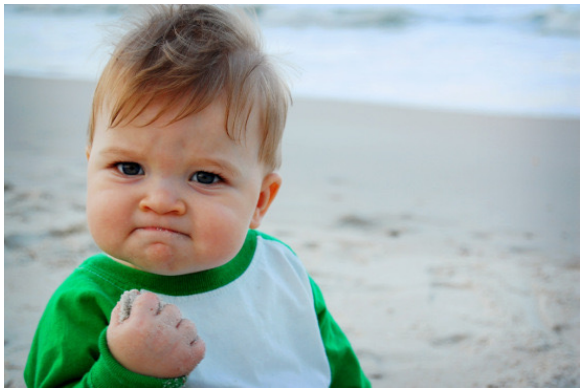





https://medium.com/@garypetercox/10-best-project-management-tools-for-streamlined-team-collaboration-a8f9c31fc62d

Project Success

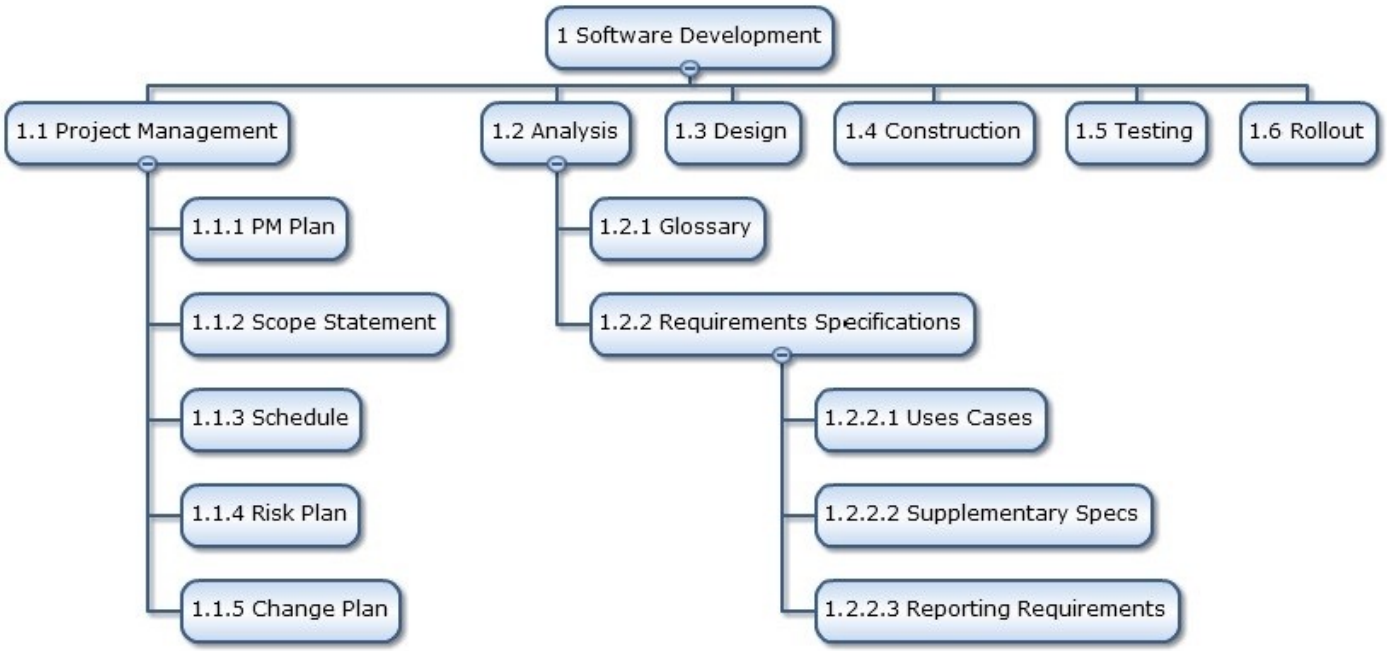
- There are several ways to define project success:
  - The project met scope, time, and cost goals
  - The project satisfied the customer / sponsor
  - The results of the project met its main objective, such as making or saving a certain amount of money, providing a good return on investment, or simply making the sponsors happy
- Top 3 success factors:
  - Adequate funding
  - Staff expertise
  - Engagement from all stakeholders



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Start Planning Your Project…

- Know your project goal, scope and deliverables
- Example of technique: **Work Breakdown Structure (WBS)**



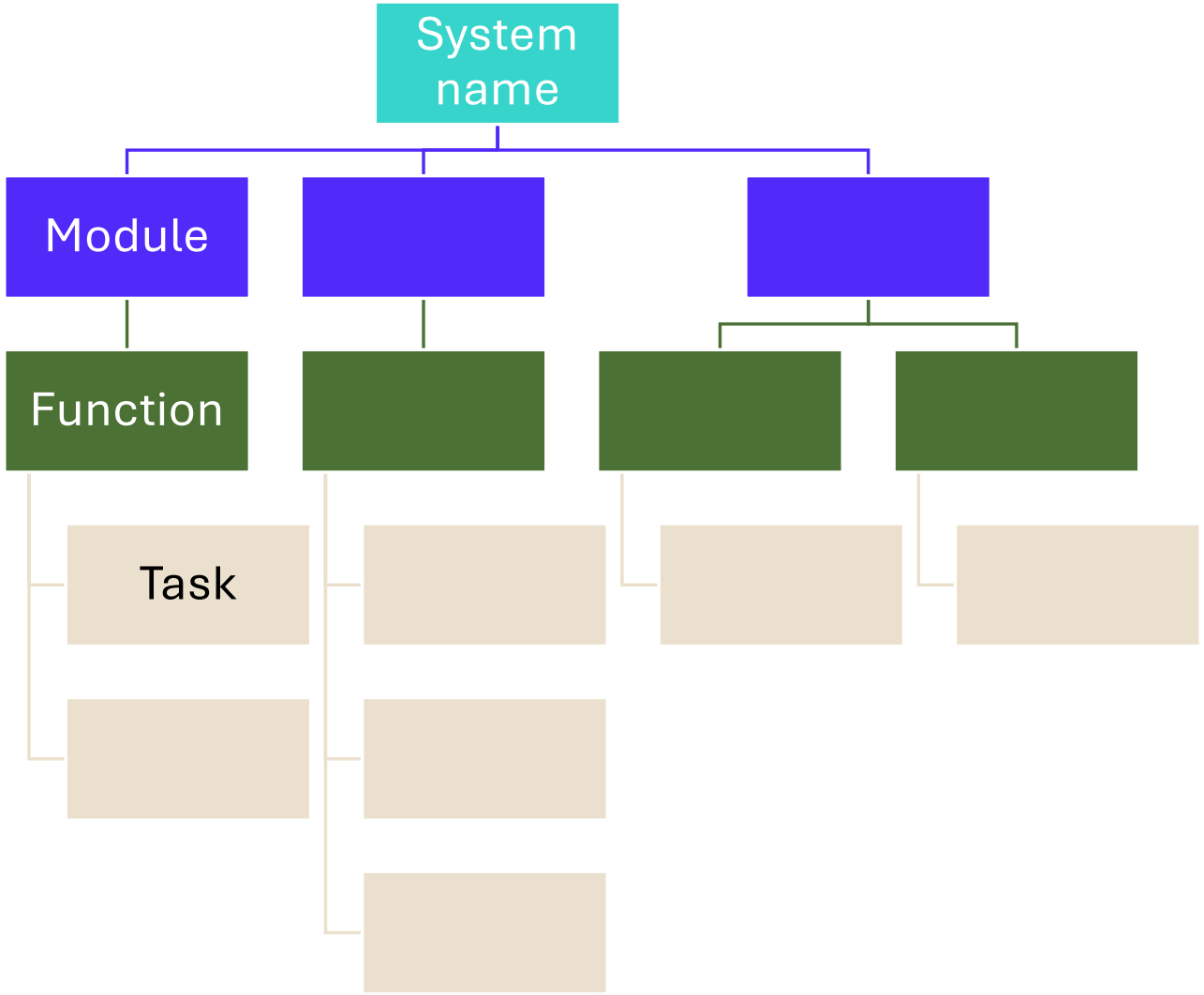
Create a WBS

- Define the project scope
  - What functions will be included?
- Identify major deliverables
  - Break the scope into **high-level components** (often aligned with software development phases or big features)
    - Project Management
    - Requirements Analysis
    - UI/UX Design
    - Development
    - Testing
    - Deployment
    - Training & Documentation
- Break down deliverables into sub-deliverables
  - Development
    - Login & registration
    - Book borrow
    - Book return
    - Fine calculation

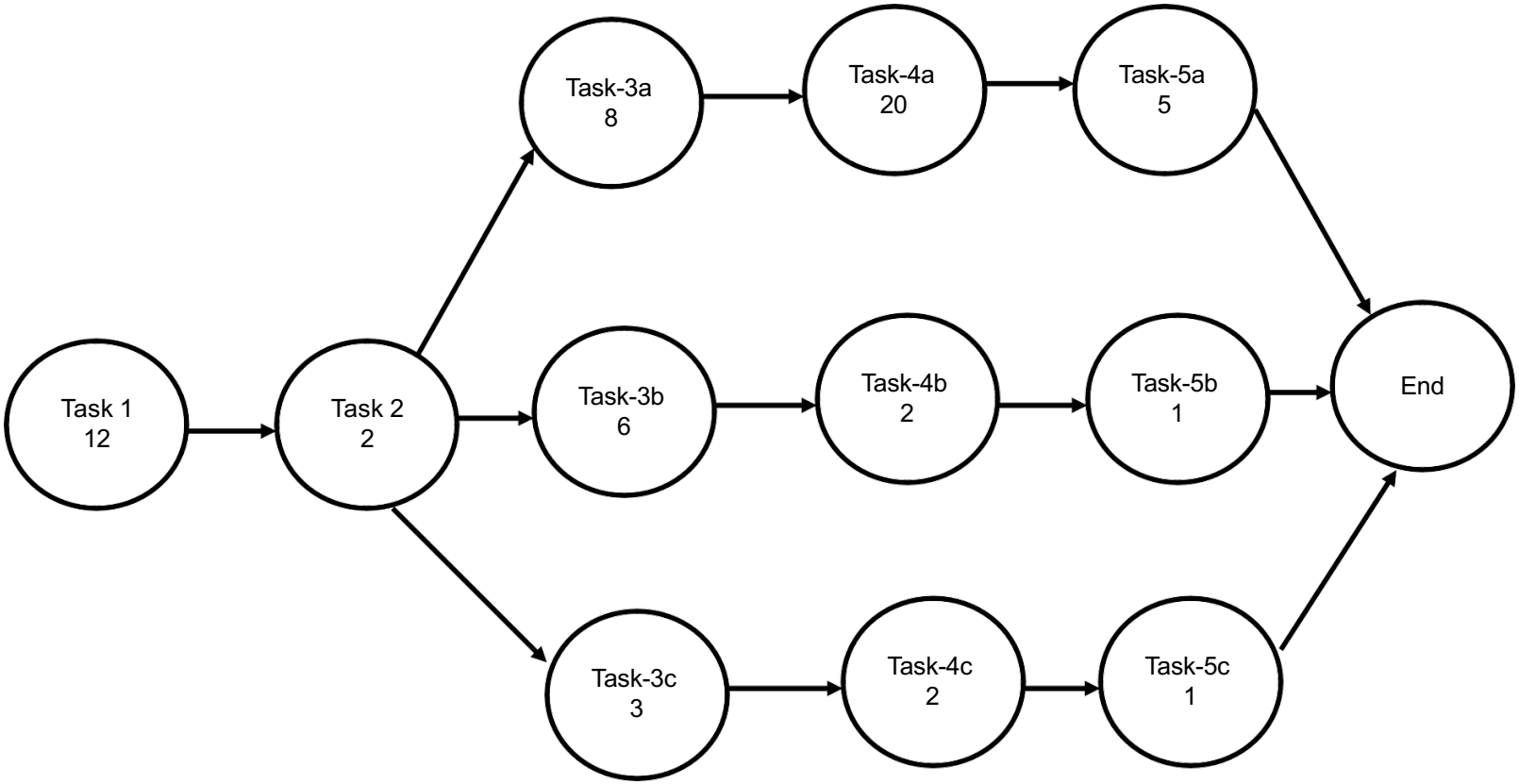
Create a WBS (Cont.)

- Sequence tasks
  - Sequential
  - Parallel
- Estimate time needed to complete each task
- Assign responsible person to each task
- Outline the timeline of the project

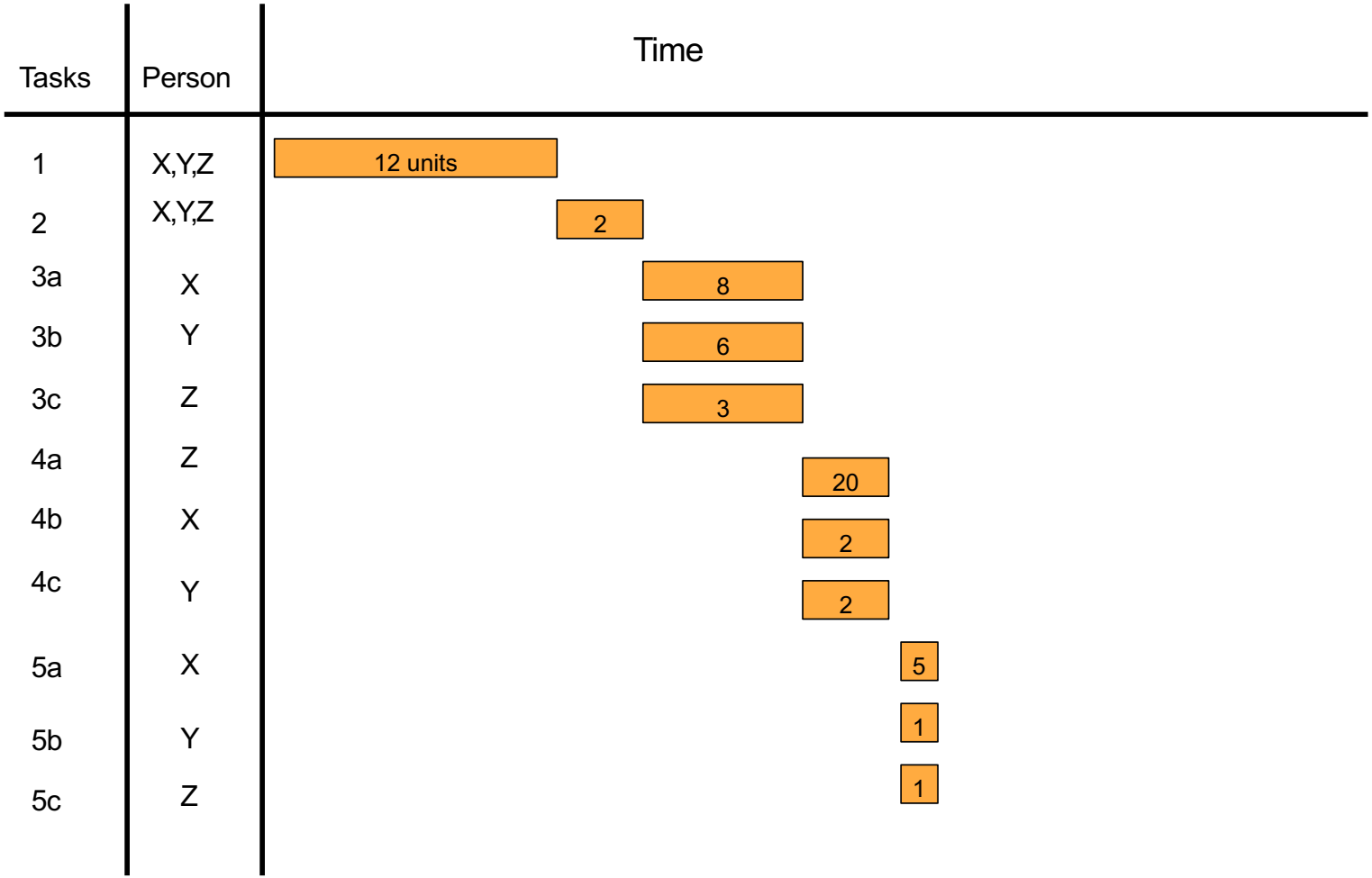
WBS



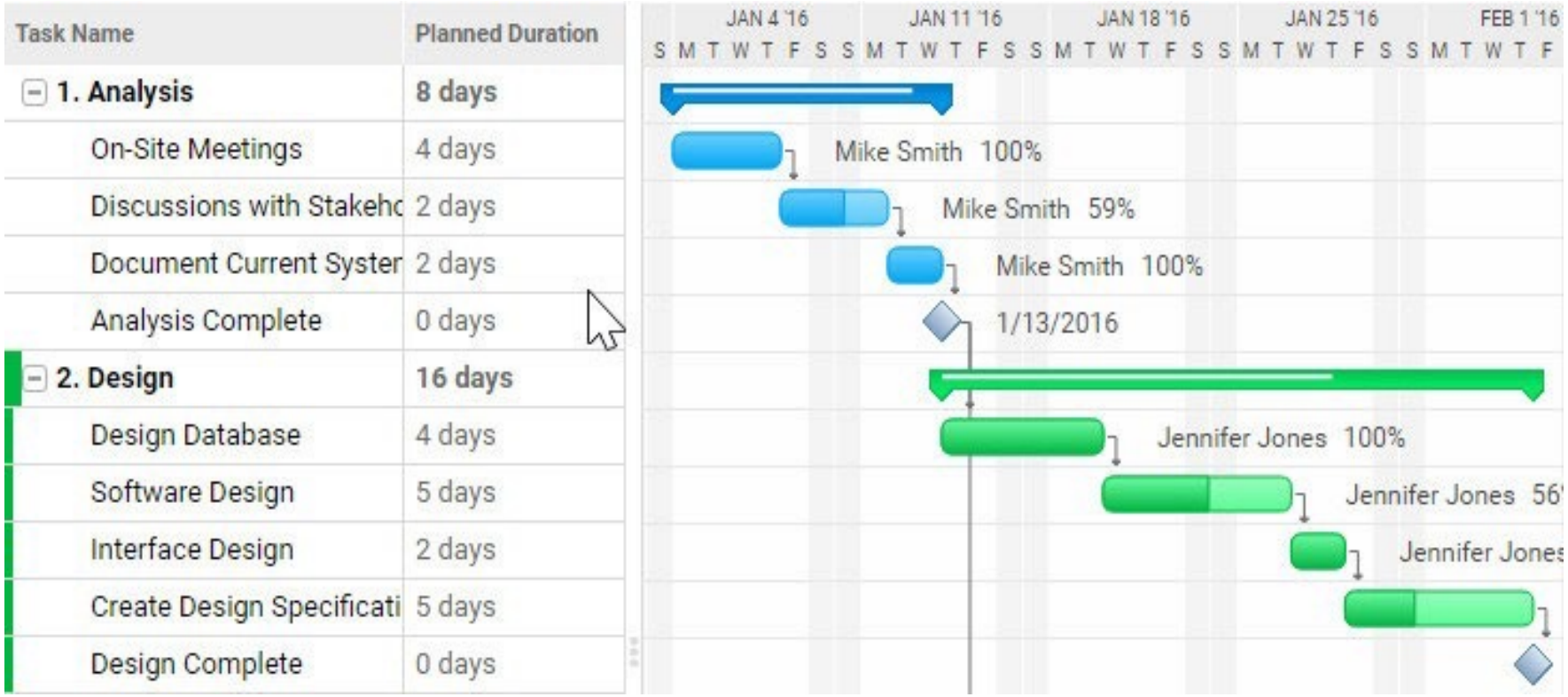
Program Evaluation and Review Technique (PERT)



End Result of WBS = Initial Schedule Estimate



Gantt Chart



https://www.projectmanager.com/gantt-chart

Working Exercise



Advantages of Project Management

- Better control of financial, physical, and human resources
- Improved customer relations
- Shorter development times
- Lower costs and improved productivity
- Higher quality and increased reliability
- Higher profit margins
- Better internal coordination
- Positive impact on meeting strategic goals
- Higher worker morale

Goals of Software Project Management

